Shaping and Stabilizing the Active Phase: The Role of Carbon Surface Defects in Carbon-Supported Co Fischer-Tropsch Catalysts

Felix Herold[1,2] *and Magnus Rønning*[2]

[1]*Friedrich-Alexander Universität Erlangen-Nürnberg, Fürth, Germany*

[2]*Norwegian University of Science and Technology, Trondheim, Norway*

Introduction

Carbon catalyst supports offer a compelling alternative to traditional oxide supports for cobalt-based Fischer-Tropsch catalysts.[1] However, identifying the optimal carbon support is challenging due to the extensive structural diversity among carbon materials, where even minor structural variations can significantly affect catalytic performance. In this work, we aim to emphasize the role of surface defects in carbon on the formation and stabilization of the active cobalt phase in cobalt-supported Fischer-Tropsch synthesis (FTS) catalysts.

Results

To investigate the impact of carbon surface defects on the performance of Co-based FTS catalysts, carbon black (CB) and CB annealed at 1900°C (CB-1900) were selected as model supports. Characterization of these materials showed that both exhibited similar specific surface areas (CB 115 m²/g, CB-1900 85 m²/g). However, they differed significantly in their nanostructural properties, such as crystallite size, graphene layer spacing, and sp³ carbon content (Fig. 2a). In this context, the non-annealed CB exhibited a high surface defect density, while annealing at 1900°C was shown to substantially reduce the number of defects.

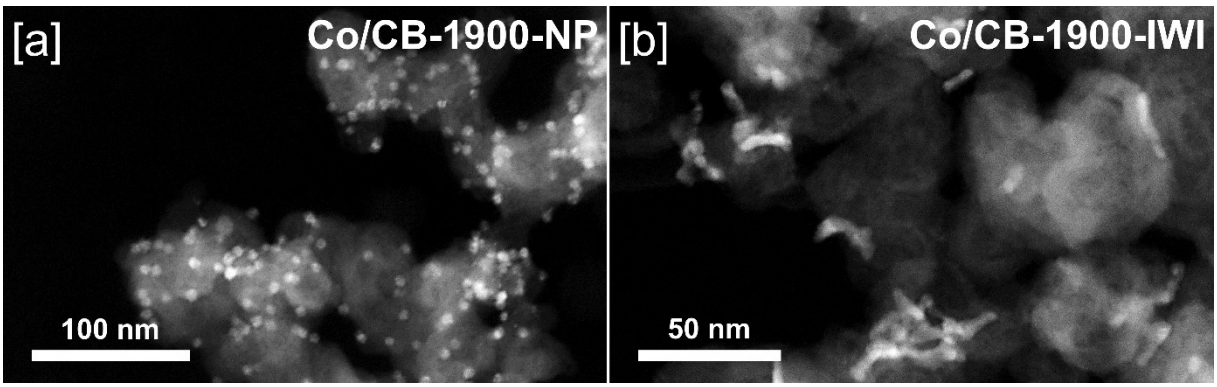

Figure 1. STEM-HAADF micrographs of pristine [a] Co/CB-1900-NP and [b] Co/CB-1900-IWI.

Catalyst synthesis was performed using both incipient wetness impregnation (IWI) and deposition of pre-synthesized colloidal cobalt nanoparticles (NP, Fig. 1). The formation and stabilization of cobalt nanoparticles were found to be strongly dependent on the carbon surface defect density. For catalysts loaded with pre-synthesized Co nanoparticles, the more defective CB support offered a higher number of anchoring sites, impeding Co particle growth during catalyst reduction and FTS compared to CB-1900, translating to higher catalytic activity (Fig. 2). For the catalysts prepared by IWI, Co dispersion was considerably higher on the defect-rich CB than on the less-defective CB-1900. As FTS is structure-sensitive, with Co nanoparticles smaller than 6-8 nm showing low intrinsic activity[2], the low average Co particle size on defective CB (3.6 ± 2.1 nm after reduction) yielded low intrinsic FTS activity (STY 0.009 s^{-1}). On the less-defective CB-1900, weaker Co/C interactions led to the formation of larger particles (8.0 ± 3.4 nm after reduction) during IWI, proving to be much better suited for high catalytic performance (STY 0.033 s^{-1}).

In conclusion, surface defects on carbon supports decisively influence the formation of Co nanoparticles and contribute to their stabilization. However, stabilization of Co nanoparticles does not necessarily improve catalytic performance due to the structure-sensitivity of the Fischer-Tropsch reaction.

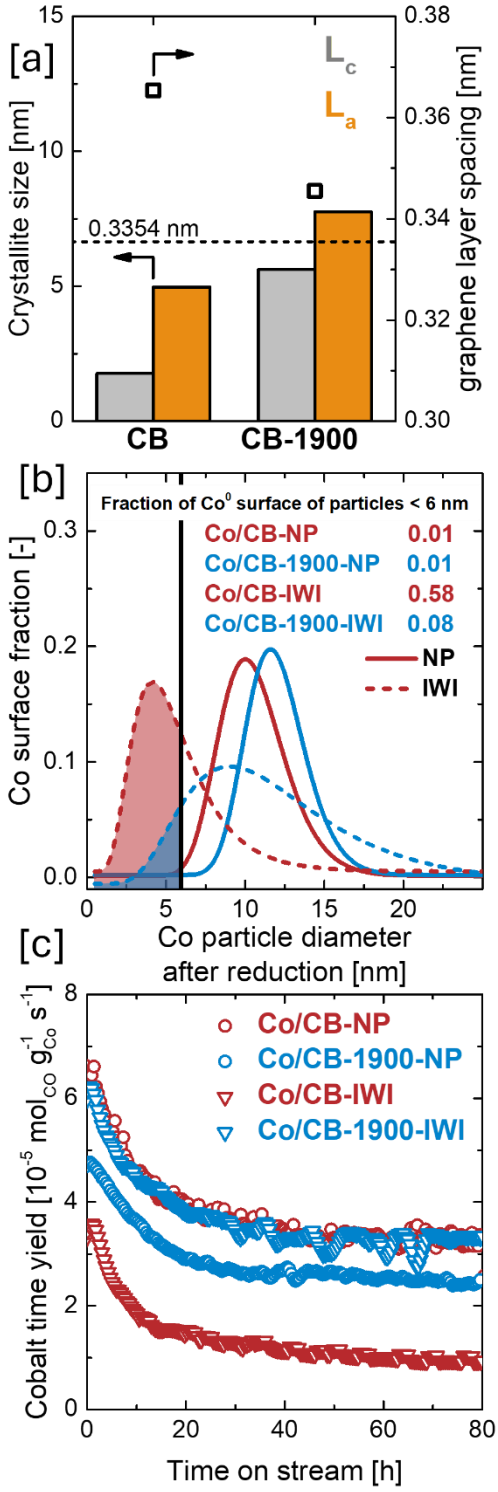

Figure 2. [a] Nanostructural parameters of CB and CB-1900. [b] Log normal fits of the surface averaged Co particle size distributions and fraction of Co surface provided by particles < 6 nm after catalyst reduction. [c] FTS performance at 220 °C, 20 bar, H_2:CO 2.1, ~20 % initial CO conversion.

References

[1] T. O. Eschemann et al. *J. Catal.* **2015**, *328*, 130.

[2] G. L. Bezemer et al. *J. Am. Chem. Soc.* **2006**, *128*, 3956.